Yibing Lu

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github.com/Yibing-L | Home+ code

Education

Cornell University, College of Engineering Jan 2025 - Dec 2025

Master of Engineering in Computer Science | Ithaca, NY

Cornell University, Department of Human Centered Design Aug 2021 - Dec 2024

Bachelor of Science in Design and Environmental Analysis

Minor in Computer Science | Ithaca, NY

Research Experience

Architectural Robotics Lab, Cornell University Feb 2025 - Present

Research Assistant

Advisors: Keith Evan Green, Tapomayukh Bhattacharjee, Haym Hirsh

Home+ — Gesture-Controlled Assistive Robotics

- Led development of the Home+ mobile assistive robot for people with upper-limb mobility limitations, integrating robot software and hardware around accessible gesture control for everyday home tasks.
- Developed URDF/Xacro models and a ROS 2 manipulation stack with MoveIt 2, KDL inverse kinematics, OMPL planning, and coordinated base-and-arm execution.
- Integrated RGB-D gesture recognition using MediaPipe hand/arm landmarks and PyTorch Attention-BiLSTM inference, connecting temporal motion features to task-level robot commands.
- Integrated multi-camera RGB-D perception, MolmoPoint/SAM2 target localization, and TF2 transforms to provide 3D manipulation targets and collision-aware scene representations.
- Developed host–Arduino closed-loop control with encoder feedback, timestamped trajectory streaming, serial protocol validation, and measured completion checks.
- Prototyped a C++ nvblox mapping adapter for CUDA-backed TSDF reconstruction and robot self-filtering, updating MoveIt with collision meshes from RGB-D observations.
- Prepared study protocols, the IRB protocol and experiment materials for evaluating gesture interaction under upper-limb movement constraints.
- Led a source-grounded retrospective of Home+ from 2009 to 2026, comparing embodiments, prototype fidelity, and research transitions to develop six lessons for long-term HRI research.

Design and Augmented Intelligence Lab, Cornell University Jun 2022 - Dec 2024

Research Assistant | Advisor: Saleh Kalantari

- Built data collection and visualization tools for indoor wayfinding research, including an interface for logging user trajectories and spatial decision points in real time.
- Implemented a trajectory analysis pipeline to extract navigation patterns, hesitation events, and route efficiency metrics from raw position data.
- Supported IRB-approved user studies, contributing to experiment design, participant coordination, and manuscript preparation.

Research Outputs

Under Review

Lu, Y., Hirsh, H., Bhattacharjee, T., Walker, I., & Green, K. E. Seventeen Years in the Making: Tracing Long-Term Design in Assistive Robotic Furnishings. *HRI 2027.* Under review.

Poster and Extended Abstract

Lu, Y., Bhattacharjee, T., Hirsh, H., Green, K. E. Home+: Gesture-to-Action Pipeline for an Assistive Robotic Table-Lamp. *Northeast Robotics Colloquium (NERC),* 2025. Poster and extended abstract.

Unpublished Manuscript

Lu, Y., Wanye, A., Yang, Q., Kalantari, S. WayFind: A Behavioral Mapping Tool for Real-World Wayfinding Trajectories and Experiences.

Technical and Research Skills

Programming and Development: Python, C++, MATLAB, Bash, Linux/Ubuntu, Git, CMake, colcon, regression testing.

Robotics and Planning: ROS 2 (rclpy/rclcpp), MoveIt 2, URDF/Xacro, TF2, RViz, OMPL, KDL inverse kinematics, collision-aware and Cartesian motion planning, mobile manipulation.

Perception and Machine Learning: PyTorch, TensorFlow, OpenCV, NumPy, MediaPipe, RGB-D and multi-camera perception, temporal gesture recognition, Attention-BiLSTM, MolmoPoint/SAM2 integration.

3D Mapping: Point-cloud processing, depth-to-3D projection, OctoMap, nvblox/TSDF integration, collision-mesh generation, robot self-filtering.

Embedded Control and Integration: Arduino Mega, PID control, magnetic/wheel encoders, wheel odometry, serial protocols/PySerial, trajectory streaming, sensor/actuator integration.

Design and Prototyping: SolidWorks, Rhino, Fusion 360, Figma, Adobe Creative Suite.

Research Methods: Human-centered design, accessibility-informed gesture design, experimental design, study protocols/IRB preparation, usability testing, trajectory analysis, data visualization, longitudinal design analysis.

Industry Experience

Nanjing Weiyan Network Technology, China Jan 2024 - Aug 2024

Software Engineer Intern

- Developed a web application with ML-based features; contributed to backend architecture and data pipelines.

Nanjing Kaijiang Technology, China May 2022 - Aug 2022

UX Design Intern

- Designed mobile interfaces through user research, prototyping, and usability testing.

Awards

Krouse Fellowship and Human Ecology Undergraduate Research Fund, Cornell (2024)

Summer Undergraduate Research Fellowship, Center for Integrative Developmental Science, Cornell (2023)

Dean's List (6 semesters)